LANGUAGE AS A CLINICAL INTERFACE: NLP-BASED ASSESSMENT, ILLNESS MONITORING, AND RESPONSIBLE AI IN SCHIZOPHRENIA

Start ≈ 0:00 · 51 words

Thank you, Professor Kishimoto. Good afternoon, everyone. I am Hironobu Nakamura, a psychiatrist at the Institute of Science Tokyo. Today I would like to talk about language as a clinical interface: how natural language processing can help us assess and monitor schizophrenia, and what it takes to use this technology responsibly.

Next: 2 · COI Disclosure

COI DISCLOSURE

Start ≈ 0:24 · 18 words

I have no conflicts of interest to disclose, and the study was approved by the relevant ethics committees.

LANGUAGE: A WINDOW INTO THE MIND

Start ≈ 0:32 · 137 words

Let me start with a simple question:
What does language mean to us?

In daily practice, whether on the ward or in the outpatient clinic, our patients' words are our richest source of information. Language is a window into our minds. It helps us recognize symptoms, make diagnoses, and understand our relationships and our humanity.

Yet for more than a century, we relied mainly on qualitative descriptions of speech: tangential, impoverished, or marked by loose associations. Natural language processing now allows us to measure these patterns.

For several years, I have been exploring this question as part of the UNDERPIN project, led by Professor Kishimoto.

Today's talk has five parts: a brief history, the basic methods, our study, a map of the field, and finally, the promise and the risks of bringing these tools into the clinic.

TWO HISTORIES CONVERGE

Start ≈ 1:35 · 122 words

This slide shows how two fields developed in parallel and gradually converged.

In psychiatry, Bleuler described the loosening of associations in 1911, and Andreasen later developed a clinician-rated scale to assess thought disorder.

In computer science, natural language processing evolved from hand-written rules to statistical methods and latent semantic analysis. Then came word embeddings such as Word2Vec in 2013, followed by the Transformer in 2017 and BERT in 2018. More recently, tools such as ChatGPT and Claude have brought large language models into everyday life.

An important point of convergence came in 2007, when latent semantic analysis was used to measure coherence in schizophrenia. Later studies used speech features to predict psychosis onset in high-risk youth. Our work builds on these developments.

Next: 5 · Four Families of NLP Models

FOUR FAMILIES OF NLP MODELS

Start ≈ 2:31 · 115 words

Next, we need to decide how to measure it. Roughly, there are four families of models. Rule- and count-based methods, such as dictionaries, parsers, and TF-IDF, tell us which words and structures are used. Static embeddings such as Word2Vec place words in a semantic space, so we can measure how close they are. Contextual models such as BERT and SentenceBERT represent meaning in context, which is useful for coherence. Finally, large language models can rate, summarise, and generate text without task-specific training. Moving to the right, representational power increases, but interpretability decreases. Because our goal was clinical understanding, we deliberately combined the first three families, so that every marker can be explained to a clinician.

TRANSITION

Start ≈ 3:24 · 7 words

Now, let’s move on to our study.

INTRODUCTION

Start ≈ 3:28 · 146 words

Language in schizophrenia has long been studied.

In recent years, natural language processing, or NLP, has enabled the quantitative analysis of linguistic abnormalities, with potential clinical applications in detection, monitoring, and prediction—such as onset, prognosis, and relapse.

Previous studies have found reduced syntactic complexity and impaired semantic coherence, along with other features such as poverty of content, lack of cohesion, use of metaphorical language, emotional valence, and pronoun use.

Several computational models have been developed, including speech graph analysis for positive formal thought disorder, semantic density models for negative thought disorder, and vector-based models such as latent semantic analysis and coherence measures.

However, most prior work has been limited by a narrow focus, small samples, and single-task designs.

To address these gaps, we conducted a large-scale, multi-feature NLP analysis, with the hypothesis that reduced syntactic complexity and increased semantic proximity represent core linguistic abnormalities in schizophrenia.

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Next: 8 · Conceptual Diagram 1: Integrative Approach ▶

CONCEPTUAL DIAGRAM 1: INTEGRATIVE APPROACH

Start ≈ 4:35 · 113 words

We adopted an integrative approach to capture linguistic abnormalities in schizophrenia across multiple dimensions and levels of language. To do this, we combined several NLP methods.

At the morphological level, we examined features such as word count, part-of-speech frequencies, and pronoun use.

At the syntactic level, we measured complexity using features such as tree depth and clause nesting depth.

At the semantic level, we used static word embeddings to examine semantic similarity between words and the density of semantic networks.

At the contextual level, we assessed sentence similarity, coherence, and continuity across discourse.

Prior studies often examined these features separately. Our approach brings them together in a comprehensive analysis of a large sample.

CONCEPTUAL DIAGRAM 2 : PAST AND PRESENT

Start ≈ 5:27 · 113 words

Another perspective of this study is the connection between past and present. Linguistic abnormalities in schizophrenia have long been investigated through classical psychopathology and linguistics, which identified well-known descriptive features.

Our study takes these classical findings as hypotheses, and examines them using natural language processing and quantitative linguistics. For example, poverty of thought can be quantified as vocabulary, the ratio of fillers and perseveration; neologisms as words absent from a dictionary; incoherence as syntactic abnormality; and loosening of associations as abnormal semantic similarity between words. This represents a transition from descriptive interpretation to comprehensive, quantitative NLP.

The ultimate goal is to bridge classical insights with modern computational tools, combining hypothesis-driven and data-driven approaches.

This study was conducted as part of the UNDERPIN project, the details of which have been published elsewhere.

Our sample consisted of 104 patients with schizophrenia and 101 healthy controls, recruited from seven hospitals and three outpatient clinics across Japan.

Participants were 20 years or older. Patients met diagnostic criteria for schizophrenia based on DSM-5 or ICD-10. Healthy controls had no psychiatric history. All participants were native Japanese speakers or fluent in Japanese.

All participants provided written informed consent, and the institutional review boards of Keio University and all participating institutions approved the study. Data were collected between 2018 and 2024.

MATERIAL & METHOD: PROCEDURES

Each participant completed between two and five visits, with at least a one-month interval between assessments.
Symptom severity in patients with schizophrenia was evaluated using the Positive and Negative Syndrome Scale, or PANSS.

Each interview lasted 30 to 60 minutes and included three tasks.
First, a free conversation, covering topics such as illness, daily life, hobbies, and recent news.
Second, a picture description task, using standardized images from the VPTA, COGNISTAT, and WAB.
Third, a storytelling task, in which participants were asked to recount the story of Cinderella, or, if unfamiliar, the Japanese folktale Kaguya-Hime.
The picture description and storytelling tasks together took about 10 minutes.

Interviewers maintained a neutral style, used minimal prompting, and both interviewer and participant wore headset microphones.
All sessions were audio-recorded for subsequent NLP analysis.

METHOD: SPEECH ANALYSES

Start ≈ 8:06 · 116 words

All speech data were transcribed into text, and only the patients' speech was analyzed, with the interviewer's speech excluded.

For feature extraction, we examined multiple levels of language.

At the morphological level, we applied part-of-speech tagging and performed Named Entity Recognition for references to time, place, person, and even mythological agents such as gods, using the GiNZA parser.

At the syntactic level, we measured dependency distance, tree depth, the depth of central embeddings, and other indices of syntactic complexity.

At the semantic level, we calculated word similarity using Word2Vec, and performed semantic network analysiscombining Word2Vec with TF-IDF.

Finally, at the contextual level, we assessed sentence similarity through cosine distance using sentenceBERT, and we also evaluated perseveration.

METHODS: ANALYSIS PIPELINE

Start ≈ 9:00 · 181 words

We conducted statistical analyses according to this pipeline, after calculating NLP features.

For feature processing, we processed 76 NLP features for each task and each visit.
To adjust for differences in speech length, features were expressed as ratios when appropriate, and then z-standardized using parameters estimated from the training set.

For exploratory analyses, we compared patients with schizophrenia and healthy controls using univariate generalized estimating equations at the subject level, averaging across visits.
These results were used for description only, and not for final feature selection.

To reduce dimensionality, we conducted exploratory factor analysis and extracted components.
From each component, we selected a representative feature based on the largest absolute loading, while checking correlations to avoid multicollinearity.

For validation, we fitted task-specific GEEs—for free conversation, picture description, and storytelling.
The outcome was schizophrenia versus control, and predictors included the representative features, along with age and sex as covariates.

Finally, we evaluated predictive performance on the held-out test set.
Performance was assessed using AUC, sensitivity, and specificity, with all preprocessing and feature selection confined to the training set to prevent information leakage.

RESULTS: DEMOGRAPHICS

Our sample included 104 patients with schizophrenia and 101 healthy controls.
The two groups were similar in age and gender distribution, with no significant differences.

As expected, patients with schizophrenia showed significantly higher PANSS scores across total, positive, negative, and general psychopathology subscales.

In terms of years of education, patients had a lower average compared with controls.

All participants completed the free conversation task, and almost all completed the picture description and storytelling tasks.

Next: 15 · Methods: ANALYSIS PIPELINE

METHODS: ANALYSIS PIPELINE

Start ≈ 10:57 · 11 words

Next, I will present the results of the exploratory group comparisons.

In free conversation, 35 of the 76 features differed significantly after FDR correction, across all levels of language. Here are two examples: patients used more person-related named entities, and fewer adverbs. These exploratory results are descriptive only. They tell us that the abnormalities are widespread, but they do not tell us which ones are essential.

RESULTS: EXPLORATORY ANALYSES (SEMANTIC NETWORK)

Similarly, in the semantic network analysis, one of the key indicators—the clustering coefficient—showed a significant group difference between patients and controls.

Moreover, the clustering coefficient was significantly correlated with PANSS negative symptom scores, as well as with the fluency subscale.

SEMANTIC NETWORK ANALYSIS USING A TF-IDF FRAMEWORK

Start ≈ 11:46 · 149 words

Let me take a moment to explain semantic network analysis briefly.

We utilized the comprehensive UNDERPIN dataset to develop a TF–IDF model, which calculates the frequency and importance of a word in a given document.
From each participant's speech, we extracted the top 10 percent of words with the highest TF–IDF scores.

We then constructed semantic graphs where content words served as nodes.
Each node was connected to up to 10 words that exceeded a similarity threshold of 0.3, using NetworkX in Python for graph creation.

Finally, we computed several network indicators: the clustering coefficient, diameter, average distance, and average closeness centrality.

Here, you see an example of these semantic graphs: on the left, a patient with severe schizophrenia, and on the right, a healthy control.
Visually, it may not be easy to distinguish the two graphs.
Therefore, I will now present the summary of our semantic network findings.

Next: 19 · Summary of Results: Semantic Network Analysis

SUMMARY OF RESULTS: SEMANTIC NETWORK ANALYSIS

Start ≈ 12:55 · 99 words

Here is the summary of the semantic network analysis.
In schizophrenia, we observed increases in the clustering coefficient and density, while average distance, diameter, and average closeness centrality decreased.
The asterisks mark the statistically significant features.

Overall, this means that in schizophrenia, semantic connections are overly tight both locally and globally. However, this kind of over-connectedness does not improve the efficiency of information transfer.

On the right, I show schematic graphs.
They may not be very clear, but they illustrate the idea that in schizophrenia, local clusters are denser, yet connections to more distant nodes are not easily accessible.

METHODS: ANALYSIS PIPELINE

Start ≈ 13:41 · 26 words

With so many features differing, we asked which dimensions are truly essential. To answer this in an interpretable way, we reduced the features with factor analysis.

RESULT: FACTOR ANALYSIS & GEE (TASK 1)

Start ≈ 13:53 · 133 words

In free conversation, factor analysis gave a stable nine-factor solution, and we kept one representative feature per factor, with low collinearity. In the multivariable GEE model, three features remained significant. First, higher mean pairwise word similarity: a one-standard-deviation increase was associated with about a 2.3-fold increase in the odds of schizophrenia. Second and third, lower ratios of case particles and adverbs. In Japanese, case particles such as ga, o, and ni mark who did what to whom. In the independent test set, the model reached an AUC of 0.87, with a sensitivity of 0.76 and a specificity of 0.90. Notably, among the major parts of speech, only adverbs were significantly reduced, not nouns, verbs, or adjectives. In Japanese, adverbs modify not only verbs but whole sentences, adding manner, degree, and the speaker's stance.

Next: 22 · Result: Comparison of Tasks

RESULT: COMPARISON OF TASKS

Start ≈ 14:54 · 92 words

We then applied the same pipeline to storytelling and picture description. Adverbial features were significant in every task, and case particles were reduced in a task-invariant way. Semantic-network and discourse features also contributed across tasks. Interestingly, perseveration was strongly elevated in storytelling, mirroring the clinical phenomenon of perseveration. Sensitivity analyses showed that the adverb effect was strongest in free conversation, whereas the reduction in case particles did not depend on the task. This suggests that case-particle reduction reflects a stable, task-general simplification, while reduced modification is most visible when speech is spontaneous.

DISCUSSION

Start ≈ 15:36 · 125 words

Taken together, we propose a tripartite linguistic phenotype of schizophrenia. First, reduced morphosyntactic explicitness: fewer case particles, so less overt marking of who did what to whom. Second, semantic narrowing: words drawn from close neighbourhoods. Third, weakened contextual modulation: fewer adverbs, which add manner, degree, and nuance. These dimensions are separable but interacting, and they go beyond the positive-negative dichotomy of thought disorder. Similar findings in Turkish, another agglutinative language, suggest shared vulnerabilities whose expression depends on language type, and the semantic findings parallel neuroimaging studies of network integration. Clinically, these markers are simple, interpretable, and related to PANSS severity, so they could serve as accessible proxies of illness burden. But these interpretations remain hypotheses that need to be tested with cognitive and functional measures.

LIMITATIONS

Start ≈ 16:34 · 101 words

There are several limitations to note.

First, our sample composition included only patients with schizophrenia, and did not include a clinical high-risk group.

Second, regarding severity, the mean PANSS score was about 75, indicating mostly mild to moderate cases.
It is possible that patients with more severe symptoms might show additional linguistic abnormalities.

Third, in terms of language and generalizability, all of our data were in Japanese, which is an agglutinative language.
Therefore, cross-linguistic validation will be necessary for broader applicability.

Finally, regarding speech modality, we analyzed spoken language.
It is possible that written data may capture semantic coherence more precisely.

TRANSITION

Start ≈ 17:21 · 15 words

Now, let’s move on to the next topics; mapping the field and promise & peril.

NLP × PSYCHIATRY: RECENT TRENDS

Start ≈ 17:28 · 109 words

Let me place our study in the context of recent trends in psychiatric NLP.

Early studies showed the potential of speech to predict psychosis onset. Research then expanded to language dimensions and mechanisms, with larger cohorts such as UNDERPIN. More recently, international collaborations have expanded, while large language models have introduced new opportunities and risks.

The current focus is on potential clinical applications, including thought-disorder assessment, symptom monitoring, and multilingual relapse prediction. Overall, the field is moving toward multidimensional, longitudinal, and multilingual approaches, with a growing emphasis on relapse prevention.

Our study contributes an interpretable, multilevel analysis of Japanese speech. Our next step is longitudinal monitoring using these features.

PROMISE AND PERIL

Start ≈ 18:18 · 179 words

I began by describing language as a window into the mind. That image captures both the promise and the risks of NLP.

On the promise side, these tools could support more objective and reproducible severity assessments, provide earlier warnings of relapse, and help us track treatment response. They could also enable monitoring between visits, reduce variability between raters, and support screening in larger populations.

But speech can reveal a person's identity and deeply private experiences. Monitoring may feel like surveillance and undermine trust, especially without meaningful consent. Models can be biased across languages, dialects, educational backgrounds, and cultures. Large language models can be difficult to interpret and may generate false information. Other concerns include overdiagnosis, labelling, and excessive trust in automated judgments. In conversation, generative AI may also echo or reinforce delusional beliefs.

Language data are intimate data. In psychiatry, they often reflect a person's fears, beliefs, and relationships. The same window that helps us understand a person can also leave them exposed. Our responsibility is to deepen understanding and improve care while protecting that person's privacy and autonomy.

Next: 28 · Toward Responsible Use

TOWARD RESPONSIBLE USE

Start ≈ 19:41 · 87 words

Responsible implementation rests on three pillars: evidence, trust, and clinical integration.

First, evidence. We need external and cross-linguistic validation, as well as longitudinal and prospective studies.

Second, trust. We must build privacy protection into the design, including through de-identification, and make the tools transparent and explainable.

Third, clinical integration. Clinicians should remain involved in decision-making, and systems must ensure that high-stakes decisions, such as diagnosis, remain in human hands.

The goal is to use AI to support and deepen clinical understanding, while clinicians retain responsibility for care.

CONCLUSION: TAKE-HOME MESSAGES

Start ≈ 20:21 · 74 words

To conclude, I have three messages. First, language is a clinical interface, and NLP can turn it into measurable, interpretable markers. Second, in Japanese speech, schizophrenia shows a tripartite phenotype, with fewer case particles, semantic narrowing, and fewer adverbs, giving strong discrimination and cross-task consistency. Third, the path to the clinic runs through validation and responsibility: AI should augment, not replace, clinical understanding, the therapeutic relationship, and human responsibility. Thank you for your attention.